

TEACH YOUR DRONES

To Do More Than Fly



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First shown at CES 2016, Etos is a BMW i8 with an auto-pilot and a drone onboard. Frank M. Rinderknecht, boss of the Swiss creative think-tank Rinspeed, intends to use the UFO-esque drone on board Etos for performing services such as picking up deliveries from stores as the car drives you home, while you relax with a book in the driver's seat. Drone applications have definitely gone way beyond their initial use as aerial torpedoes, unmanned weapons platforms and surveillance.

Financial benefits look very promising for developers of well-engineered drones. In a recent *Forbes* article by Baldwin Cunningham, it was said that the drone economy could be as incredible as the app economy we had seen in the last decade. To engineer drones that can handle next-generation applications, let us take a look at the technologies available for you.

Propulsion

Before you try to put smarts into your drone, you need to get your drone off the ground. When you select your mix of motor and propeller, you need to make sure that there is enough thrust to comfortably pick up the entire drone.

As an example, a motor and propeller combination that delivers 500 grams of thrust in a quadcopter configuration would be able to barely lift a two-kilogram drone. This is not an ideal

situation and so it is recommended that the weight of your drone be less than half the maximum thrust your configuration is capable of delivering.

Direct current (DC) brushless motors are the most commonly used ones here. The Kv rating you find in these motors signify revolutions per minute per volt. A motor rated at 1000Kv will spin 1000 times per volt when there is no load attached to it. This is just a theoretical value, so it is not recommended to test your brushless motor without a load on it. To control speed, you can increase or decrease the voltage applied to the motor.

Electronic speed controllers (ESCs) are devices that let you control revolutions per minute (rpm) of the motor reliably. These devices are able to handle the maximum current the motor might consume at the exact voltage as required by the user. Most motor manufacturers offer their own ESCs.

Some of the popular motors in the market are the ones by Lynxmotion, TMotor, DJI and Storm. You need not look at motor brands; you should be fine as long as the specifications match your rpm and thrust requirements. Some expensive motors we came across come with better ball bearings that promise longer life and more reliability. For example, MN3508 motor built for aerial-photography drones comes with ball bearings that are twice the standard size seen.

However, electric motors come with an inherent weakness in the form of range and flight duration due to battery-pack limitations. That is where people have gotten creative to repurpose tried-and-tested technologies like internal combustion engines. Yair is one such product that comes with a ten cubic centimetre combustion engine paired with running engine control. The result? A drone with a range of over 55 kilometres!

